

Group Contribution Log

Machine Learning Approach for Optimal Football XI Selection Based on Opponent Tactical Analysis

Course: Machine Learning Research Project | Section: J | Group: 02 | AIUB

Class Serial	Contributor Name	Student ID	Sections/Task	Specific Contributions
04	Md. Sifti Al Mahmud	23-50099-1	- Dataset Construction & Cleaning (Section 4.1) - Feature Engineering & Leakage Audit (Section 4.3) - Time-Based Split & SMOTE Implementation (Section 4.4) - XGBoost Model Development & Tuning (Section 4.5)	- Scraped and merged 6 FBref tables; created Match_ID; reconciled team name variants - Engineered 178 features across 5 groups; applied correlation filter to reduce to 126 - Implemented chronological 80/20 split; applied SMOTE only to training fold - Tuned XGBoost hyperparameters; implemented scale_pos_weight for class imbalance - Conducted TimeSeriesSplit 3-fold cross-validation; ran Wilcoxon signed-rank test - Generated SHAP beeswarm and gain-based feature importance plots
38	Md. Towhidul Islam	23-55036-3	- Custom Position-Weighted Match Rating (Section 4.2) - Win Probability Model (Section 4.7) - Deployment Pipeline & Formation Search (Section 4.7 / 5.5) - Discussion & Conclusion (Sections 6 & 8)	- Designed position-weighted raw score formulas for GK, DF, MF, FW - Implemented Z-score normalisation per (Match_ID, Team, Broad_Position) group - Applied exponential decay ($\lambda=0.1$) to compute current form rating R_p - Built second XGBoost win probability classifier on 1,598 match-team observations - Implemented 8-formation search with slot-aware tactical position filling

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				(GK→CB→FB→CM→A M→FW) 12. Wrote El Clásico deployment example; authored Discussion and Conclusion sections
39	Anika Tabassum	23-55070-3	• Introduction & Literature Review (Sections 1 & 2) • Research Gap Analysis (Section 3) • Results — Calibration & Graphs (Section 5.3) • Limitations & Future Work (Section 7)	13. Wrote Introduction covering sports analytics market context and motivation 14. Reviewed and summarised 10+ related works on match prediction and player rating 15. Identified and documented the 4 research gaps addressed by this paper 16. Described calibration (reliability) curves and Brier score comparison across models 17. Compiled and formatted References section (30 citations, ACM style) 18. Wrote Limitations & Future Work section covering league generalisation and injury data gaps
40	Abrar Kabir	23-55095-3	• Model Benchmarking — Decision Tree, MLP, RF, Stacking (Section 4.5) • Evaluation Protocol & Metrics (Section 4.6) • Results — Lineup Classification & Cross-Validation (Sections 5.1 & 5.2) • SHAP Analysis & Feature Importance (Section 5.4)	19. Implemented and tuned Decision Tree (GridSearchCV), MLP (3-layer, early stopping), Random Forest (600 trees) 20. Built Stacking ensemble combining RF + XGBoost + LightGBM with Logistic Regression meta-learner 21. Defined evaluation metrics: AUC-ROC, F1-Macro, F1-Weighted, Accuracy, Brier Score 22. Generated ROC curves, confusion matrix, and model comparison tables (Table 1 & 2) 23. Computed SHAP values on 1,500-row

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				validation subset; interpreted beeswarm plot patterns 24. Prepared all evaluation figures and wrote Section 5 results narrative

Submission Date: 2nd May 2026